MIL-H-3982C

25 June 1969

SUPERSEDING
SEE 6.3

MILITARY SPECIFICATION

HARDWARE (FASTENERS AND RELATED ITEMS), PACKAGING AND PACKING FOR SHIPMENT AND STORAGE OF

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers cleaning, preserving, packaging, packing, and marking of hardware and related items classified as bolts, screws, pins, rivets, keys, nuts, washers, hinges, brackets, door buttons, knobs, blind hardware, storm sash, screen hangers, sash pulleys, and miscellaneous other fasteners and components, both metallic and nonmetallic. (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL	
NN-K-231	Kegs, Wood, Slack.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-576	Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-591	Boxes, Fiberboard, Wood-Cleated.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665	Boxes, Paperboard, Metal-Stayed (Including Stay Material).
PPP-B-676	Boxes, Set-Up, Paperboard.
PPP-C-55	Cans, Fiberboard and Paperboard, (With Paper and Metal Ends).

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MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-117	Bags, Interior Packaging.
MIL-B-121	Barrier Material, Greaseproofed, Waterproof, Flexible.
MIL-L-3150	Lubricating Oil, Preservative, Medium.
MIL-P-3420	Packaging Materials, Volatile Corrosion Inhibitor, Treated, Opaque.
MIL-I-8574	Inhibitors, Corrosion, Volatile, Utilization of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water - Vapor Proof or Waterproof Flexible.
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold Application.
MIL-B-43014	Boxes, Water Resistant Paperboard, Folding, Set-Up and Metal-Stayed.
MIL-C-3955	Cans, Fiber, Spirally Wound.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois, 60606.)

American Trucking Association Inc.
National Motor Freight Classification

Application for copies should be addressed to the American Trucking Association Inc., 1616 P Street, N.W., Washington, D.C. 20036.)

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3. REQUIREMENTS

3.1 General. Hardware items shall be prepared for shipment in accordance with the requirements of this specification. Related items of hardware not specifically covered herein shall be prepared for shipment in accordance with instructions provided for similar items.

3.1.1 Materials. Materials used in preparation for shipment shall be as specified herein. Materials not covered by specifications shall be of good comercial quality, suitable for the purposes for which intended, and shall be free of defects and imperfections that will affect adversely the serviceability of the packaged item.

3.2 Preservatives. Preservatives used shall be the types designated in MIL-P-116, which are listed below:

TYPE	SPECIFICATION NO.
P-1	MIL-C-16173, grade 1
P-2	MIL-C-16173, grade 2
P-7	MIL-L-3150
P-18	MIL-P-3420
P-19	MIL-P-16173, grade A

3.3 Preservative application and packaging.

3.3.1 Level A.

3.3.1.1 Cleaning. Hardware items to which a preservative compound will be applied (see 3.3.1.3) shall be cleaned in accordance with cleaning process C-1 in MIL-P-116. Hardware and nonmetallic items not requiring application of a preservative shall also be cleaned in accordance with C-1.

3.3.1.2 Drying. Unless otherwise specified, required drying shall be accomplished by one or more of the procedures specified in MIL-P-116, D-1 through D-5, provided the drying procedure is not injurious to the item.

3.3.1.3 Preservatives and preservative application. Preservatives specified herein shall conform to the applicable specifications listed in 3.2 and shall be applied in accordance with MIL-P-116.

3.3.1.4 Unit packaging.

3.3.1.4.1 Quantity. Unless otherwise specified in the contract or order (see 6.2), identical items shall be unit packaged in numerical quantities specified in Tables I through X. When weight and size of items are so great the interior containers are impracticable, the items are so indicated by an asterisk (*). Items marked by an asterisk shall be placed directly into exterior containers as specified in 3.4 and in quantities not exceeding weight limitations imposed by 3.4.

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3.3.1.4.2 Wrapping.

3.3.1.4.2.1 Preserved with P-1 or P-19. Items preserved with P-1 or P-19 shall be given no additional packaging until the preservative compound has hardened. After hardening of the P-1 or P-19 preservative, items so preserved may be placed directly into the exterior container with no further wrapping.

3.3.1.4.2.2 Preserved with P-2 or P-7. Unit package quantities of items coated with P-2 or P-7 shall be wrapped in greaseproof barrier material conforming to Grade A of MIL-B-121 and placed in the unit container selected. Items preserved with P-7 shall be drained of excess preservative before wrapping. Items preserved with P-2 or P-7 shall not be wrapped when the unit container is completely lined with greaseproof barrier material. Free movement of items shall be prevented by use of snug fitting containers and suitable cushioning or blocking.

3.3.1.4.2.3 Preserved with P-18. When quantities of an item are to be preserved with P-18, whether in a unit package or in an exterior container without interior packaging, the P-18 preservative shall be applied in the form of a complete wrap without the use of any other preservative and in accordance with MIL-I-8574. Supplemental oil finish on phosphate coated items does not have to be removed.

3.3.1.4.2.4 Unpreserved items. Items not requiring preservatives require no greaseproof wrapping before being placed into containers. However, they shall be wrapped or cushioned with non-corrosive material when necessary to prevent movement within the container.

3.3.1.4.3 Packaging methods.

3.3.1.4.3.1 Preserved items. Unless otherwise specified in Tables I through X or in the contract or order (see 6.2), preserved hardware items shall be packaged in accordance with Method I of MIL-P-116.

3.3.1.4.3.2 Unpreserved items. Unless otherwise specified in Tables I through X or in the contract or order (see 6.2), unpreserved hardware items shall be packaged in accordance with Method III of MIL-P-116.

3.3.1.4.4 Containers.

3.3.1.4.4.1 For unit packages with contents weighing one pound or less. Containers shall conform to one of the following: MIL-B-117 or MIL-B-43014. When an outside barrier bag is utilized, the following containers may also apply: PPP-B-566, PPP-B-665, PPP-B-676, and PPP-C-55, Type III. The use of bags shall be restricted to the packaging of small or flat items. Items having sharp edges or protrusions shall be wrapped or cushioned prior to being bagged in order to prevent damage to the bag.

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3.3.1.4.4.2 For unit packages with contents weighing more than one pound. When specified (see 6.2) containers shall conform to weather-resistant class of PPP-B-636. Unless otherwise specified, container shall conform to one of the following: MIL-C-3955 or MIL-B-43014. When an outside barrier bag is utilized the following containers may also apply: PPP-B-566, PPP-B-676, and PPP-C-55, Type III. Gross weight of unit package shall not exceed 20 pounds for PPP-B-636, PPP-B-665, and MIL-B-43014 containers, and 10 pounds for other specification containers cited.

3.3.1.4.4.3 For hardware assortments. Unless otherwise specified, like items within a hardware assortment shall be packaged together and identified, using either compartmented containers or individual inner packages.

3.3.1.5 Intermediate packaging. Intermediate packaging is required when the container for the unit package is a bag. Unless otherwise specified, (see 6.2), intermediate packaging is required for boxed unit containers which are 15 cubic inches or less in size. Intermediate packaging shall be accomplished by placing a quantity of unit packages into containers conforming to PPP-B-676, PPP-B-665, PPP-B-636 or MIL-B-43014. Gross weight shall not exceed 10 pounds for PPP-B-676 containers and 20 pounds for PPP-B-665, PPP-B-636 and MIL-B-43014 containers. Intermediate package shall provide a snug fit for contents. Quantity of unit packages to an intermediate container shall be uniform in any shipment, except final intermediate container may hold a lesser quantity. Unit packages exceeding either 10 pounds gross weight or 15 cubic inches in size need not be intermediate packaged. Intermediate containers shall be closed as specified in the appendices or notes of the applicable container specifications. Army shipments shall be packaged in a weather resistant intermediate container when the boxed unit package is less than 64 cubic inches. Intermediate containers shall be limited to a maximum of 100 unit packages with a maximum net load of 40 pounds, and a maximum size of 1.5 cubic feet.

3.3.2 Level C. Hardware items shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. Packaging may conform to supplier's commercial practice when such meets the requirements of this level. Packaged quantities shall be as specified in the contract or order (see 6.2).

3.4 Packing. Packing shall be Level A, B, or C, as specified (see 6.2). Contents of a shipping container shall be material of only one stock number, except the packing of material of more than one stock number is permitted under conditions specified in 3.4.7.

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3.4.1 Level A. Unless otherwise specified, packaged hardware shall be packed in boxes of the weather resistant or overseas class or type conforming to one of the following: PPP-B-576, PPP-B-585, PPP-B-601, or PPP-B-621. When specified (see 6.2), plywood boxes shall be surface-treated in accordance with the requirements of PPP-B-601. Strapping and closure shall be in accordance with the appendix to the applicable container specification. Boxes shall be constructed for type 2 loads when such construction is included in the box specification, except that PPP-B-585 boxes shall be for type 3 loads. Unpackaged hardware items shall be packed into containers conforming to one of the following: NN-K-231, PPP-B-601, overseas type and PPP-B-621, class 2. Boxes for unpackaged hardware items shall be constructed for type 3 loads when such construction is included in the box specification selected. Unless otherwise specified in the contract or order, (see 6.2), the gross weight of kegs and wood or wood-cleated boxes shall not exceed 200 pounds. If a gross weight exceeding 200 pounds is authorized by the contract or order for boxes used for shipping containers, boxes shall be provided with skids. However, gross weight of multiple packed items (see 3.4.7) shall not exceed 200 pounds. Closure and strapping shall conform to the box specification and the appendix thereto.

3.4.1.1 Case liners for level A shipment. When specified (see 6.2) containers shall be lined with a case liner conforming to MIL-L-10547 and sealed according to the appendix of that specification, with the exception that no liner shall be required when a shipping container is packed with packages or items meeting the following conditions:

a. Unpackaged items which are completely painted and have no unprotected interior surfaces.

b. Unpackaged items which are completely coated with type P-1 or P-19 preservative conforming to MIL-P-116.

3.4.2 Level B. Packaged hardware items shall be packed in domestic type or class boxes conforming to PPP-B-585, PPP-B-601, or PPP-B-621 or in weather resistant type or class boxes conforming to PPP-B-591, PPP-B-636, and PPP-B-640. Closure shall be in accordance with the appendix to the applicable specification. When fiberboard containers are utilized, the fiberboard shall have a minimum Millen Test of 275 pounds. Boxes shall be constructed for type 2 loads when such construction is included in the box specification, except that PPP-B-585 boxes shall be for type 3 loads. Unpackaged hardware items shall be packed into containers conforming to one of the following specifications: NN-K-231; PPP-B-601, domestic type; PPP-B-621, class 1. Boxes for unpackaged hardware items shall be constructed for type 3 loads when such construction is included in the box specification selected. Unless otherwise specified in the contract or order, (see 6.2), the gross weight of kegs and wood or wood-cleated boxes shall not exceed 200 pounds, and the gross weight of fiberboard boxes shall not exceed the limitations of the fiberboard box specification. If a gross weight exceeding 200 pounds is authorized by the contract or order for boxes used for shipping containers, boxes shall be provided with skids. Gross weight of multiple

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packed items (see 3.4.7) shall not exceed 200 pounds. Box closure and strapping shall conform to the box specification and the appendix thereto.

3.4.3 Level C. Hardware items shall be packed in shipping containers of type, size, and kind commonly used for the purpose in a manner which will insure acceptance by common carrier and safe delivery at destination. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

3.4.3.1 Unit or intermediate packages as shipping containers. When the total quantity of an item to a single destination in the continental United States is not more than the intermediate package quantity, material may be shipped in commercial containers conforming to Parcel Post or common carrier regulations, as applicable to the mode of shipment. Unit or intermediate containers conforming to these requirements may be shipped without further packing. In such cases shipping containers shall be marked for level C.

3.4.4 Bulk shipment. When specified in the contract or order (see 6.2), hardware items of like size and description shall be bulked shipped, regardless of package quantities specified in this specification. When bulk shipment is authorized, only exterior containers specified for unpackaged items in 3.4.1, 3.4.2, and 3.4.3 shall be used.

3.4.4.1 Shipment of component parts. In bulk shipment of fasteners, component parts such as washers, nuts and pins, shall not be furnished loose in the shipping container. The required quantity of such parts shall be packaged and the package enclosed with the bulk shipment. At the option of the supplier, fasteners with nuts may be furnished with the nuts either assembled to the fastener or packaged as described above. The option does not apply when the procurement specification requires the nuts to be assembled to the fastener.

3.4.5 Blocking and cushioning. Interior packages or items placed directly into the exterior containers shall be blocked or cushioned in accordance with MIL-STD-1186.

3.4.6 Size of shipping containers. For the purpose of conserving shipping and storage space, shipping containers shall have a minimum cubic displacement based on overall length, width, and depth, compatible with the requirements for protection of the items contained.

3.4.7 Multiple packed shipping containers. Multiple packing is permitted when all material of one stock number will not fill one container to capacity. Within a multiple-packed shipping container, packages of identical stock numbers shall be cartonized or wrapped together as a means of seperating the material into groups for easy handling and identification.

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3.4.8 Unitized loads. When specified (see 6.2) for Army procurement, maximum use of unitized loads shall be utilized (CONEX containers, palletized unit loads, and consolidation containers) whenever the size, weight, shape, compatability and quantity of packages or packs permit.

3.5 Marking. Interior packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129. Markings on interior packages shall include contract or order number. Unpackaged items require no interior marking.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government, The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all filled containers on the same contract or order, which are prepared for shipment under essentially the same conditions and are submitted for inspection at the same time and place.

4.3 Sampling for inspection.

4.3.1 For examination. Samples for examination shall be selected at random from each inspection lot in accordance with procedures prescribed in MIL-STD-105 for inspection level II.

4.3.2 For testing. Samples for testing shall be selected at random from each inspection lot in accordance with procedures prescribed in MIL-STD-105 for Inspection Level S-4 for normal inspection.

4.4 Acceptable quality levels.

4.4.1 Examination. The acceptable quality level for examination shall be 2.5 percent defective for all defects.

4.4.2 Tests. The acceptable quality level for tests shall be 4.0 percent defective.

4.5 Quality conformance inspection.

4.5.1 Examination. Each of the sample shipping containers or interior packages selected in accordance with 4.4.1 shall be examined to verify compliance with this specification. Any shipping container or package in the sample containing one or more defects shall be rejected,

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and if the number of defective shipping containers or packages in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5.1.1 Classification of defects. Examination of shipping containers and interior packages shall be conducted as follows:

Categories	Defects	Gage or other method of inspection
Critical	None defined	
Major	AQL 2.5 percent	
101	Incorrect preservative applied, preservative applied when not required, or omitted when required (3.3.1.3)	Visual
102	Improper method of unit packaging utilized (3.3.1.4.3)	Visual
103	Improper type, grade, class, or size of unit package used (3.3.1.4.4)	Visual
104	Failure of unit package	4.5.2
105	Defective or damaged packaging material used (3.3.1)	Visual
106	Markings and identification on interior packages illegible, missing or incorrect (3.5)	Visual
107	Excessive net or gross wieght of interior packages as applicable (3.3.1.4 and 3.3.1.5)	Weighing
108	Case liner missing, damaged or incorrectly constructed (3.4.1.1)	Visual
109	Unauthorized shipping container used (3.4.1, 3.4.2, and 3.4.3)	Visual
110	Incorrect type, class, or grade of shipping container used (3.4.1, 3.4.2, and 3.4.3)	Visual and Measure
111	Improper closure of shipping container including strapping, when required (3.4.1, 3.4.2, and 3.4.3)	Visual
112	Excessive shipping container size with respect to contents (3.4.6)	Visual
113	Multiple packed items not intermediate packaged (3.4.7)	Visual
114	Markings and identification on shipping container illegible, missing or incorrect (3.5)	Visual
115	Contents of shipping container improperly cushioned or blocked (3.4.5)	Visual
116	Excessive gross weight of shipping container (3.4)	Weighing
117	Skids omitted (3.4.1 and 3.4.2)	Visual
Minor	None defined	

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4.5.2 Tests.

4.5.2.1 Unit package. Tests for unit packages selected in accordance with 4.3.2 shall be conducted as prescribed in MIL-P-116 for the applicable packaging method.

5. PREPARATION FOR DELIVERY. Not applicable to this specification.

6. NOTES.

6.1 Intended use. This specification is intended to be used for direct reference in contracts or orders and for reference in Section 5 of commodity specifications and for use in preparation of packaging data sheets for the preservation, packaging, and packing of Hardware (Fasteners and Related Items).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Level of packaging and level of packing required (see 3.3 and 3.4).

c. Quantity per unit package, if other than as provided in this specification (see 3.3.1.4.1).

d. Unit packages if other than specified (see 3.3.1.4.4.2).

e. If intermediate packaging is not required (see 3.3.1.5).

f. Quantity per Level C, unit package, if required (see 3.3.2).

g. Surface treatment for plywood boxes, if required (see 3.4.1).

h. Case liners, if required (see 3.4.1.1).

j. If bulk shipment is required (see 3.4.4).

k. Weight limitation of shipping containers, if required to be greater than provided in this specification (see 3.4.1 and 3.4.2).

l. Unitized loads, when specified (see 3.4.8).

6.3 Supersession data. This specification supersedes MIL-H-3982B Amendment 1, 28 February 1968 and MIL-H-22173, Hardware, Builders, Packaging and Packing of, dated 7 August 1959.

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Custodians:
Army - WC
Navy - OS
Air Force - 82

Review activities:
Army - SM, AT, AV, ME, MI
Navy - SA, YD, SH, AS
Air Force - None
GSA
DSA-IS

User activities:
Army - None
Navy - MC, EC

Preparing activity:
Army - WC

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TABLE I. Bolts, Lag; Bolts, Machine; Studs; Setscrews; Screws, Cap. (Includes items with nuts, washers, etc.)

(Numbers indicate the quantity of items to be placed in a unit package; asterisks (*) indicate sizes to be packed directly into the exterior container without interior packages. When the application of a preservative coating is required, items shall be coated with type P-2 or P-7 preservative (see 3.3.1.3.1). As an alternative, items may be protected by type P-18 (see 3.3.1.4.2.3).)

Nominal Diameter (Inches)		Nominal Length (Inches)												
Fraction	Decimal	0 To 1	Over 1 To 2	Over 2 To 3	Over 3 To 4	Over 4 To 5	Over 5 To 6	Over 6 To 7	Over 7 To 8	Over 8 To 9	Over 9 To 10	Over 10 To 11	Over 11 To 12	Over 12
1/16		100												
	.075	100												
	.086	100	100	100										
	.100	100	100	100										
	.112	100	100	100	100									A
1/8		100	100	100	100									L
	.140	100	100	100	100									L
	.154	100	100	100	100									
	.164	100	100	100	100									D
	.182	100	100	100	100									I
3/16		100	100	100	100	100	100	100	100	50	50	50	50	A
	.190	100	100	100	100	100	100	100	100	50	50	50	50	M
	.216	100	100	100	100	100	100	100	100	50	50	50	50	E
7/32		100	100	100	100	100	100	100	100	50	50	50	50	T
1/4		100	100	100	100	100	100	100	100	50	50	50	50	E
5/16		100	100	100	50	50	50	50	50	25	25	25	25	R
3/8		100	100	50	50	50	50	50	50	25	25	25	25	S
7/16		100	50	50	50	25	25	25	25	25	25	25	25	
1/2		50	50	50	25	25	25	25	25	25	25	25	25	
9/16		50	50	25	25	25	25	25	25	*	*	*	*	*
5/8		25	25	25	25	25	25	25	25	*	*	*	*	
3/4		20	20	20	20	20	20	20	20	*	*	*	*	
7/8		15	15	15	15	15	15	15	15	*	*	*	*	
1		10	10	10	10	10	10	10	10	*	*	*	*	
1-1/8			*	*	*	*	*	*	*	*	*	*	*	
1-1/4			*	*	*	*	*	*	*	*	*	*	*	

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TABLE II. Pins, Cotter; Pins Straight; Pins, Grooved,; pins, Shoulder; Pins, Spring; Pins, Tapered.

(Numbers indicate the quantity of items to be placed in a unit package. When application of a preservative coating is required, items shall be coated with type P-2 or P-7 (see 3.3.1.3.1). As an alternative, items may be protected by type P-18 (see 3.3.1.4.2.3).)

Nominal Diameter (Inches)		Nominal Length (Inches)												
Fraction	Decimal	To 1	Over 1 To 1-1/4	Over 1-1/4 To 2	Over 2 To 3	Over 3 To 3-1/4	Over 3-1/4 To 4	Over 4 To 4-1/2	Over 4-1/2 To 5	Over 5 To 5-1/2	Over 5-1/2 To 6	Over 6 To 6-1/2	Over 6-1/2 To 7	Over 7 to 8
1/64	.0156	100												
	.027	100												
1/32	.0312	100	100	100	100									
	.042	100	100	100										
3/64	.0468	100	100	100										
	.056	100												
1/16	.0625	100	100	100	100	100	100	100	100					
5/64	.078	100	100	100	100									
	.087	100	100	100	100	100	100	100	100	100	100	100	100	100
3/32	.093	100	100	100	100	100	100	100	100	100	100	100	100	100
7/64	.109	100	100	100	100	100	100	100	100	100	100	100	100	100
1/8	.125	100	100	100	100	100	100	100	100	100	100	100	100	100
	.133	100	100	100	100	100	100							
9/64	.140	100	100	100										
5/32	.156	100	100	100	100	100	100							
11/64	.171	100	100	100										
3/16	.187	100	100	100	100	100	100	100	100	100	100	50	50	50
13/64	.203	100	100	100										
7/32	.218	100	100	100	100	100	100	100	100	100	50	50	50	50
1/4	.250	100	100	100	100	100	100	100	100	50	50	50	25	25
9/32	.281	100	100	100										
5/16	.312	100	100	100	100	100	100	100	50	50	25	25	25	25
11/32	.343	100	100	100	50	50	50							
3/8	.375	100	100	50	50	50	50	25	25	25	25	10	10	10
13/32	.406	100	100	50	50									
7/16	.437	100	50	50	50	50	50	25	25	10	10			
1/2	.500	50	50	50	50	50	50	25	25	10	10	10	10	10
9/16	.563	50	50	50	25	25	25	10	10	10	10	10	10	10
19/32	.593	50	25	25	25	25	10	10	10					
5/8	.625	50	25	25	25	10	10	10	10	10	10	10	10	10
11/16	.656	50	25	25	25	10	10	10	10	10	10	10	10	10
23/32	.718	25	25	25	10	10	10	10	10	10	10	10	10	5
3/4	.750	25	10	10	10	10	10	10	10	10	10	5	5	5
7/8	.875	25	10	10	10	10	10	10	10	10	10	5	5	5
15/16	.937	25	10	10	10	10	10	10	10	10	10	5	5	5
63/64	.984	10	10	10	10	10	10	10	10	10	10	5	5	5
1	1.0	10	10	10	10	10	10	10	10	10	10	5	5	5
OVER 1 INCH--All lengths: One Per Package.														

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TABLE III. Rivets, Solid; Rivets, Split; Rivets, Tubular; and Pins, Rivet.

(Numbers indicate the quantity of items to be placed in a unit package; asterisks (*) indicate sizes to be packed directly into the exterior container without interior packages. When application of a preservative coating is required, items shall be coated with type P-2 or P-7 preservative (see 3.3.1.3.1). As an alternative, items may be protected by type P-18 (see 3.3.1.4.2.3). When specified, rivets shall be packaged in one pound, or greater, unit packages).

Nominal Diameter Inches		Nominal Length (Inches)										
Fraction	Decimal	To 1/2	Over 1/2 To 1	Over 1 To 1-1/2	Over 1-1/2 To 1-3/4	Over 1-3/4 To 2	Over 2 To 2-1/2	Over 2-1/2 To 3	Over 3 To 3-1/2	Over 3-1/2 To 4	Over 4 To 5	Over 5 To 6
3/64	.048	1000										
1/16	.063	1000										
3/32	.093	1000	500	250	250	250						
7/64	.109	500	250	250								
1/8	.125	500	250	100	100	100	100	100				
9/64	.141	250	100									
5/32	.156	250	100	100	100	100	100	100				
3/16	.187	250	100	100	100	100	100	100	100	100	100	100
7/32	.218	100	100	100	100	100	100					
1/4	.250	100	100	100	100	100	100	100	50	50	50	50
9/32	.281	100	100	100	100	100						
5/16	.312	100	100	100	100	50	50	50	50	50	*	*
3/8	.375	100	100	50	50	50	50	25	*	*	*	*
13/32	.406	100	50	50	50	25	25	25	*	*	*	*
7/16	.437	100	50	50	25	25	25	25	*	*	*	*
1/2	.500	50	25	25	25	25	25	25	*	*	*	*
5/8	.625	25	25	25	25	25	*	*	*	*	*	*
3/4	.750											
AND OVER - ALL LENGTHS *												

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TABLE IV. Screws, Drive; Screws, Assembled Washer; Screws Machine; Screws Tapping; Screws, Wood; Thumbscrews.

(Numbers indicate the quantity of items to be placed in a unit package. When the application of a preservative coating is required, items shall be coated with type P-2 or P-7 preservative (see 3.3.1.3.1). As an alternative, items may be protected by type P-18 (see 3.3.1.4.2.3).)

SIZE		NOMINAL LENGTH (INCHES)							
No.	Nominal Diameter (Inches)	0 To 1	Over 1 To 1-3/4	Over 1-3/4 To 2-1/2	Over 2-1/2 To 2-3/4	Over 2-3/4 To 3	Over 3 To 4	Over 4 To 4-1/2	Over 4-1/2 To 6
0	.060	100							
1	.073	100							
2	.086	100							
3	.099	100	100						
4	.112	100	100						
5	.122	100	100	100					
5	.125	100	100	100					
6	.137	100	100	100					
6	.138	100	100	100					
7	.151	100	100	100	100				
7	.155	100	100	100	100				
8	.163	100	100	100	100	100	100		
8	.164	100	100	100	100	100	100		
8	.165	100	100	100	100	100	100		
9	.177	100	100	100	100	100	100		
10	.186	100	100	100	100	100	100		
10	.190	100	100	100	100	100	100		
10	.191	100	100	100	100	100	100		
11	.203	100	100	100	100	100	100		
12	.212	100	100	100	100	100	100	100	
12	.216	100	100	100	100	100	100	100	
12	.218	100	100	100	100	100	100	100	
14	.242	100	100	100	100	100	100	50	50
14	.243	100	100	100	100	100	100	50	50
	1/4	100	100	100	100	100	100	50	50
14	.251	100	100	100	100	100	100	50	50
16	.268	100	100	100	100	100	100	50	50
18	.294		100	100	100	100	100	50	50
	5/16		100	100	100	100	100	50	50
20	.320		100	100	100	100	100	50	50
22	.346		100	100	100	100	50	50	50
24	.372		100	100	100	50	50	50	50

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TABLE V. Keys, Woodruff; Keys, Machine; and Keys, Gib.

(Numbers indicate the quantity of items to be placed in a unit package; asterisks (*) indicate sizes to be packed directly into the exterior container without interior packages. When application of a preservative coating is required, items shall be coated with type P-2 or P-7 preservative (see 3.3.1.3.1). As an alternative, items may be protected by type P-18 (see 3.3.1.4.2.3).)

Nominal Width (Inches)	Nominal Length (Inches)											
	0 To 3/8	Over 3/8 To 1/2	Over 1/2 To 3/4	Over 3/4 To 1	Over 1 To 1-1/2	Over 1-1/2 To 1-3/4	Over 1-3/4 To 2	Over 2 To 2-/12	Over 2-1/2 To 3	Over 3 To 4	Over 4 To 5	Over 5 To 6
1/16	100	100	100	100	100	100	100	100	100	100	100	100
3/32	100	100	100	100	100	100	100	100	100	100	100	100
1/8	100	100	100	100	100	100	100	100	100	100	100	100
5/32	100	100	100	100	100	100	100	100	100	100	100	100
3/16	100	100	100	100	100	100	100	100	100	100	50	50
7/32	100	100	100	100	100	100	100	100	100	100	50	50
1/4	100	100	100	100	100	100	100	100	100	50	50	50
5/16		100	100	100	100	100	100	100	50	50	50	25
3/8		100	100	100	100	50	50	50	50	25	25	25
7/16				100	100	50	50	50	50	25	25	25
1/2			100	100	100	50	50	50	25	25	25	25
5/8			100	100	50	50	50	25	25	25	25	10
3/4				50	50	50	50	25	25	25	10	10
7/8		50	50	50	25	25	25	25	10	10	10	10
1					25	25	25	10	10	10	10	*
1-1/8						25	10	10	10	10	*	*

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TABLE VI. Nuts.

(Numbers indicate the quantity of items to be placed in a unit package; asterisks (*) indicate sizes to be packed directly into the exterior container without interior packages. When application of a preservative coating is required, items shall be coated with type P-2 or P-7 preservative (see 3.3.1.3.1). As an alternative, items may be protected by type P-18 (see 3.3.1.4.2.3).)

Bolt or Screw Size	Nominal Thread Diameter (Inches)	Nuts, Square, (Full and Jam Thickness)	Nuts, Hex. Plain, Slotted and Castellated, (Full and Jam Thickness)	Nuts, Hex, Lock		Nuts, Machine Screw Hex. and Square	Nuts				
				Heavy	Light		CAP	PLATE	STAMPED	WELDING	WING
1	.073					100					
2	.086					100			100		
3	.099					100			100		
4	.112					100			100		100
5	.125					100			100		100
6	.138					100	100	100	100		100
8	.164					100	100	100	100		100
10	.190			100	100	100	100	100	100		100
12	.212			100	100	100	100	100	100		100
1/4	.250	100	100	100	100	100	100	100	100	100	100
5/16	.313	100	100	100	100	100	100	100	100	100	100
3/8	.375	100	100	100	100	100	100	100	100	100	100
7/16	.438	50	50	100	100		100	100	100	50	50
1/2	.500	50	50	100	100		100	100	100	50	50
9/16	.563	50	50	50	50		50		100	50	50
5/8	.625	25	25	50	50		50		100	25	25
3/4	.750	20	20	50	50		50		100		25
7/8	.875	15	15	25	25		25		100		
1	1.0	10	10	25	25		25		50		
1-1/8	1.125	*	*	20	20				50		
1-1/4	1.250	*	*	20	20				50		
1-3/8	1.375	*	*	*	20				50		
1-1/2	1.500	*	*	*	10				25		
1-5/8	1.625	*	*	*					25		
1-3/4	1.750	*	*	*					25		
1-7/8	1.875	*	*	*					25		
2	2.0	*	*	*					10		
2-1/4	2.250	*	*	*					10		
2-1/2	2.500	*	*	*					10		

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TABLE VII. Washers

(Numbers indicate the quantity of items to be placed in a unit package; asterisks (*) indicate sizes to be packed directly into the exterior container without interior packages. When application of a preservative coating is required, items shall be coated with type P-2 or P-7 preservative (see 3.3.1.3.1). As an alternative, items may be protected by type P-18 (see 3.3.1.4.2.3). When other preservation instruction appears in a column of this table, that instruction shall govern. When specified, washers shall be packaged in 1, 5, 10 or 25-pound packages.)

Bolt or Screw Size	Nominal Hole Diameter (Inches)	WASHERS, FLAT ROUND SHAPE			WASHERS, FLAT SQUARE	WASHERS, LOCK			WASHERS				
		Machine; Standard; SAE	Rivet Burrs	Beveled; Shouldered	Beveled Standard	SPLIT	TOOTHED	SPRING	CLINCH	FINISHING	NON-METALLIC [1]	REDUCING	KEY
0	.070	1000						50	100	100	100	100	100
1	.083	1000						50	100	100	100	100	100
2	.093	1000	1000			1000	1000	50	100	100	100	100	100
3	.112	1000	1000			1000	1000	50	100	100	100	100	100
4	.122	1000	1000			1000	1000	50	100	100	100	100	100
1/8	.138	1000	1000		100	1000	1000	50	100	100	100	100	100
5	.140	1000	1000		100	1000	1000	50	100	100	100	100	100
6	.156	1000	500	500	100	1000	1000	50	100	100	100	100	100
8	.187	1000	500	500	100	1000	1000	50	100	100	100	100	100
9	.190	1000	500	500	100	1000	1000	50	100	100	100	100	100
10	.212	1000	500	500	100	1000	1000	50	100	100	100	100	100
12	.250	1000	500	250	100	1000	1000	25	100	100	100	100	50
14	.265	1000	500	250	100	1000	1000	25	100	100	100	100	50
1/4	.281	1000	500	250	100	1000	1000	25	100	100	100	100	50
	.312	1000	500	250	100	1000	1000	25	100	100	100	100	50
5/16	.343	1000	500	250	100	1000	1000	10	100	100	100	100	50
	.375	1000	500	250	100	1000	500	10	100	100	100	100	50
3/8	.437	1000	100	100	50	500	500	10	100	100	100	100	25
	.463	500	100	100	50	500	500	10	100	100	100	100	25
7/16	.500	500	100	100	50	250	250	10	50	50	50	50	25
1/2	.562	250	100	100	50	250	250	10	50	50	50	50	25
9/16	.625	250	50	50	50	100	100	10	50	50	50	50	25
5/8	.656	250	50	50	50	100	100	5	50	50	50	50	25
	.687	250	50	50	50	100	100	5	50	50	50	50	25
	.750	250	50	50	50	100	100	5	50	50	50	50	25
3/4	.812	100	50	50	25	100	100	5	50	50	50	50	25
	.875	100	50	50	25	100	100	5	50	50	50	50	25
7/8	.937	50	50	50	10	100	100	5	50	50	50	50	25
	1.0	50	50	50	10	100	50	5	50	50	50	50	25
1	1.125	*	*	*	10	50	50	5	25	25	25	25	25
	1.250	*	*	*	10	50	50	5	25	25	25	25	25
1-1/4	1.375	*	*	*	*	50	50	5	25	25	25	25	25
	1.500	*	*	*	*	50	50	5	25	25	25	25	25
1-1/2	1.625	*	*	*	*	*	*	5	25	25	25	25	25
	1.750	*	*	*	*	*	*	*	25	25	25	25	25
1-7/8	2.0	*	*	*	*	*	*	*	25	25	25	25	25
2	2.125	*	*	*	*	*	*	*		10	10	*	25
2-1/4	2.375	*	*	*	*	*	*	*		10	10	*	25
2-1/2	2.625	*	*	*	*	*	*	*		10	10	*	10
3	3.250	*	*	*	*	*	*	*					

[1] Method IC-1 (MIL-P-116) applies without use of a preservative.

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TABLE VIII. Fasteners.

(Numbers indicate the quantity of items to be placed in a unit package. When application of a preservative coating is required, items shall be coated with type P-2 or P-7 (see 3.3.1.3.1). As an alternative, items may be protected by type P-18 (see 3.3.1.4.2.3).)

Nominal Dimension (Diameters, lengths, etc.) (Inches)	Insert, Screw Thread[4]	Cap, Rivet[1]	Post, Snap Fastener[2]	Fastener, Spring Tension Trim[3]	Grommet, Metallic	Clinch Plate, Push Button Fastener; Clinch Plate, Snap Fastener; Clinch Plate, Turnbutton Fastener[4]	Socket, Snap Fastener[1]	Stud, Snap Fastener	Cap, Snap Fastener[1]
#0 (0.60 TO- 1/4 (.250)	100								
Over 1/4 TO 3/8 (.375)	50	100	100	100	100	100	100	100	100
Over 3/8 TO 5/8 (.625)	25								
Over 5/8	10								

1 Nominal dimension represents the outside diameter
2 Nominal dimension represents the length
3 Nominal dimension represents the hole diameter
4 Nominal dimension represents the inside diameter

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TABLE IX. HINGE, BUTT; HINGE, BUTT, SPRING; HINGE, BUTT, RING; HINGE, BUTT, OFFSET; HINGE, CABINET; HINGE, TEE; HINGE, STRAP; HASPS; HOOKS; LOCK, DOOR; LOCK, CABINET; CATCH, CUPBOARD; CLOSURE, DOOR; BRACKETS.

(NUMBERS INDICATE THE QUANTITY OF ITEMS TO BE PLACED IN A UNIT PACKAGE; ASTERISKS (*) INDICATE SIZES TO BE PACKAGED ONE (1) TO A UNIT CONTAINER, EXCEPT THAT ITEMS EXCEEDING 4 POUNDS IN WEIGHT SHALL BE PACKED DIRECTLY INTO EXTERIOR CONTAINERS WITHOUT INTERIOR PACKAGES. WHEN THE APPLICATION OF A PRESERVATIVE COATING IS REQUIRED, ITEMS SHALL BE COATED WITH TYPE P-2 OR P-7 PRESERVATIVE (SEE 3.3.1.3.1). AS AN ALTERNATIVE, ITEMS MAY BE PROTECTED BY TYPE P-18 (SEE 3.3.1.4.2.3).

NOMINAL WIDTH (INCHES)		NOMINAL LENGTH (INCHES)						
OVER	TO	TO 1	OVER 1 TO 1 1/4	OVER 1 1/4 TO 2	OVER 2 TO 3	OVER 3 TO 4	OVER 4 TO 5	OVER 5
0	1/2	10	10	10	10	10	10	
1/2	1	10	10	10	10	10	10	
1	1 1/2	10	10	10	10	10	10	
1 1/2	2	10	10	10	10	10	5	ALL
2	2 1/2	10	10	10	10		5	
2 1/2	3	10	10	10	10	5	5	WIDTHS
3	3 1/2	10	10	10	5	5	5	
3 1/2	4	10	5	5	*	*	*	*
4		*	*	*	*	*	*	

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TABLE X. KNOBS AND PULLS; BLIND HARDWARE; STORM SASH AND SCREEN HANGERS; HANGERS; PULLEYS; PULLEYS, SASH; SCUTTLE AND SKYLITE OPERATORS; BUTTONS, DOOR.

(NUMBERS INDICATE THE QUANTITY OF ITEMS TO BE PLACED IN A UNIT PACKAGE; ASTERISKS (*) INDICATE SIZES TO BE PACKAGED ONE (1) TO A UNIT CONTAINER, EXCEPT THAT ITEMS EXCEEDING 4 POUNDS IN WEIGHT SHALL BE PACKED DIRECTLY INTO EXTERIOR CONTAINERS WITHOUT INTERIOR PACKAGES. WHEN THE APPLICATION OF A PRESERVATIVE COATING IS REQUIRED, ITEMS SHALL BE COATED WITH TYPE P-2 OR P-7 PRESERVATIVE (SEE 3.3.1.3.1). AS AN ALTERNATIVE, ITEMS MAY BE PROTECTED BY TYPE P-L8 (SEE 3.3.1.4.2.3).

NOMINAL WIDTH (INCHES)		NOMINAL LENGTH (INCHES)						
OVER	TO	TO 1	OVER 1 TO 1½	OVER 1½ TO 2	OVER 2 TO 3	OVER 3 TO 3½	OVER 3½ TO 4	OVER 4
	½	10	10	10	10	10	10	
½	1	10	10	10	10	10	10	
1	1½	10	10	10	10	10	10	
1½	2	10	10	5	5	5	5	ALL
2	2½	10	5	5	5	5	5	
2½	3	10	5	*	*	*	*	WIDTHS
3	3½	10	5	*	*	*	*	
3½	4	10	5	*	*	*	*	
4		*	*	*	*	*	*	*

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